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# COMMERCIAL ITEM DESCRIPTION

## Blood Recovery Delivery System

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a blood recovery delivery system suitable for salvaging shed blood, processing and reinfusing it back in the patient, in a sterile closed system.

Salient characteristics:

General requirements. In addition to the requirements specified in this document, the requirements of MIL-D-42048 shall apply. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

Materials. The Blood Recovery Delivery System shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion-resistant. Where painting or plating is used, appropriate cleaning and priming shall be performed prior to finishing. Unless otherwise specified, finish shall be equivalent or superior to that of comparable commercial equipment, and shall be free of gaps, scratches, chips, and other imperfections. All materials used shall be suitable in every respect for the purpose intended.

All plastic components that come in contact with blood shall meet the requirements of Class IV plastic when tested in accordance with the USP.

AMSC N/A FSC 6515

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Operation. The blood recovery delivery system shall salvage blood shed during an operation or as a result of trauma. The system shall process this blood by suctioning and mixing with anticoagulent, filtering out macro-debris, centrifuging to remove plasma, other fluids and clotting factors, washing cells with normal saline, and packing cells to be suspended in saline to a reinfusion bag for return to the patient. The processing of the blood shall be accomplished in a sterile closed system automatically controlled by an internal dual microprocessor. The end product shall be of the highest quality.

The operating system shall primarily consist of a control and LED digital display panel and an operating deck.

Control and display panel. The system shall be capable of operating in either a fully automatic or manual mode. All controls and display panels shall be located on the cover of the cabinet.

An "on/off" power switch located on the front panel shall be able to turn the power on or off.

The display panel shall provide important information regarding the functioning of the system. As statistics change, they will be updated on the display.

During normal operation, the display shall provide both general information regarding the status of the system, and specific information regarding pump speed, centrifuge speed, and number of cycles in progress as a ratio to the number of cycles programmed.

The control panel shall be divided into two sections - One for operating the system in AUTO mode, and the other containing controls for MANUAL mode and other special functions.

A "HELP" switch shall be provided on the control panel. Upon powering up the system, the operator may press the "HELP" switch which will cause a list of setup instructions to be presented on the display panel. The display instructions shall provide useful instructions to the operator of the steps involved in attaching the disposable components and in reprogramming the system.



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Auto mode: The system shall be capable of operating in a fully automatic mode with minimum operator's help. The machine shall be preprogrammed to perform a protocol that will apply in most surgical applications. However, it will be possible to reprogram the parameters of this protocol, either for one procedure or permanently.

The auto mode shall primarily consist of the following functions.

A. Start Auto Process: This button shall initiate the first cycle of a set of cycles, and shall place the system in the automatic mode of operation. Pressing this button in manual mode will return the system to the automatic mode in the same cycle.

B. Pause/Resume. This shall be a two function switch. If the machine is running, pressing the PAUSE shall cause pump to halt, centrifuge to idle, and all clamps to close. If no further action is taken, centrifuge will brake to a halt. When the machine is paused or stopped, pressing RESUME shall restart the pump and centrifuge. The machine shall remember what action it was performing when it paused. When it resumes operation, it shall return to the action it was performing when paused. If the machine was in the manual mode when paused, pressing RESUME shall place the machine in the automatic mode.

C. Final cycle. This button shall be used to offer the operator four options:

1. Concentrate (bring washed red blood cells from reinfusion bag back into the bowl)
2. Proceed to wash
3. Push STOP and wait for more blood loss, or
4. Push STOP and discard. The FINAL CYCLE button shall be only functional when either the reservoir is empty or the machine has stopped or paused.

D. STOP. This button shall stop the pump and centrifuge and close and clamps. The machine shall remember which action it was performing when "STOP" was pressed.

In addition to the auto mode, the system shall include the following controls:

a. Programming. These buttons shall allow the operator to alter the automatic operation of the system by changing the program that controls operation.



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b. Wash select. These buttons shall allow the operator to choose between FILL, HALF or NO wash.

c. Pump speed. Pump speed slew buttons shall allow smooth increase or decrease of the pump speed in the range 0 to 1000 ml/m. Changes made by the pump speed shall affect the present cycle only.

d. Manual processing. In the manual mode, the operator shall be able to regulate FILL, CONCENTRATE, WASH, EMPTY, RETURN and STOP actions.

e. Special functions. These functions include REMOVE AIR, PUMP ONLY, and A and B. REMOVE AIR shall override the air detector, open the clamp, and use the pump to draw air out of reinfusion bag. PUMP ONLY shall give the operator control over the direction of the pump, and A and B shall be used with pump only to respond to display menu choices.

Operating deck. The operating deck shall primarily consist of three clamps, pump, air detector and the centrifuge well.

Clamps: The clamps shall be color-coded which shall be capable of occluding the three color-coded lines of the harness. In the automatic mode, the opening and closing of the clamps shall be activated by the machine as programmed. The functions of each clamp shall be as follows:

Red: Opens the pathway to the blood source.
Yellow: Opens the pathway to wash solution.
Blue: Opens the pathway to reinfusion bag.

Pump: Fluids shall be moved in and out of the bowl by a computer controlled reversible roller pump. At its maximum speed, it shall deliver a flow of 1000 ml/min. The pump shall be accurate to $\pm$ 5 RPM at flow rates of 0 to 1000 ml/min.

Air Detector: The system shall ultrasonically detect the air in the pump tubing leading to the centrifuge bowl. During automatic mode, the air detector shall be active all the time.

Centrifuge Well: The centrifuge well shall consist of chuck, the support arms, the well light and the sealed optics subsystem with synchronous modulation detection electronics. When the red blood cells reach a predetermined level within a bowl, the optical sensor shall advance the machine from FILL to WASH or concentrate to wash. In addition, the optical detector shall prevent red blood cells spillover into the waste bag during WASH.



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The centrifuge shall be computer controlled capable of continuous operation while maintaining accuracy of ± 5 RPM. Speed shall be adjustable from 3000 to 6000 RPM. Computer controlled centrifuge braking shall minimize braking time while preventing red cell damage. Centrifuge cover shall have an interlock to prevent opening when centrifuge is operating.

Construction. The system shall consist of a cabinet, cover, wheels, waste bag weigher, IV pole, power cord bracket, telescoping handle, and autotransfusion reservoir shell.

The cabinet and the cover shall contain all electronics hardware. This hardware shall be easily accessible for servicing. The cabinet shall contain a visible air filter. The air filter shall be reusable. The cover shall be able to fold to protect operating deck and control panel during shipping and when not in use. The cabinet shall have as a minimum eight inch rear wheels with integral latches and dual two inch front swivel casters.

Back of the cabinet shall contain an adjustable handle with cord storage bracket. The handle shall contain a telescoping pole to provide mounting position for reservoir, reinfusion bag and saline bags. The handle shall be padded and shall function as a sturdy bumper in transit. The unit shall not exceed the following dimensions:

cover open: 17"W X 29"L X 43"H

cover closed: 17"W X 29"L X 31"H

Weight of the unit shall not exceed 265 lbs.

Handle and wheel system shall allow two types of movements, conventional cart motion using front and rear wheels or hand truck motion by tilting. Two additional fixed handles - one front and one rear - shall be provided that would allow lifting the entire unit.

Side of the cabinet shall contain a waste bag weigher with hooks on which to hang the waste bag. The waste bag weigher shall check the weight of the fluid in the waste bag and alert the operator when it is almost full and also when it is completely full.



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The system shall include a reusable autotransfusion reservoir shell. The reservoir shell shall be made of polycarbonate or equivalent durable transparent plastic. The reservoir shall be composed of two sections each approximately 13 inches by 17 inches. These sections shall be connected by a metal hinge to allow smooth opening and closing. Two clamps or similar device shall be provided to make the reservoir air tight in a closed position. The overall shape and design shall be suitable for holding a 3 liter capacity autotransfusion liner.

Performance. When the blood is processed using the following protocol, the end product hematocrit shall be 50% and the processing time shall not exceed 3 min. + 10%.

| | |
|---|---|
| Incoming hematocrit | 0.35 |
| Fill speed (ml/min.) | 500.0 |
| Wash speed (ml/min.) | 600.0 |
| Empty speed (ml/min.) | 500.0 |
| Spin volume (ml) | 250.0 |
| Wash volume (ml) | 1000.0 |

The system shall perform without any degradation in performance under the following environmental conditions:

Temperature: 10 to 27 degrees C.(50 to 80 degrees F)
Humidity: 95%, noncondensing

Alternate design, construction or material. Variations in design, construction or material may be permissible provided the system contains all the features as specified, system performs as required, and such variations are acceptable by the services.

Workmanship. The blood recovery delivery system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One blood recovery delivery system, as specified, constitutes one unit.



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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. The unit of product for test purposes shall be one blood recovery delivery system.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.



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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

| NSN | ITEM IDENTIFICATION |
|---|---|
| 6515-01-156-9786 | BLOOD RECOVERY DELIVERY SYSTEM |

Ordering data. (Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order).



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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PGC 52011

Project No. 6515-4360



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